The goal of this task was to implement a **Bidirectional Long Short-Term Memory (BiLSTM)** model to enhance sequential data understanding by capturing **context from both past and future directions**.
This approach was chosen to improve accuracy and contextual understanding over traditional (unidirectional) LSTM models.

Steps Followed

1. Data Preparation

- Created a small dataset of text sentences with binary sentiment labels (1 = positive, 0 = negative).
- Used **Keras Tokenizer** to convert text into numerical sequences.
- Applied **padding** to ensure all input sequences have equal length.
- Finalized input (X) and output (y) arrays for model training.

2. Model Design

Constructed a **Bidirectional LSTM architecture** using TensorFlow/Keras:

- **Embedding layer:** Converts each word into a dense numerical vector representation.
- **Bidirectional LSTM layer:** Processes the sequence **both forward and backward**, capturing full sentence context.
- **Dropout layer:** Prevents overfitting by randomly dropping neurons during training.
- **Dense layers:** Perform non-linear transformations for final classification.
- **Output layer:** A single neuron with a sigmoid activation for binary output.

3. Model Compilation

- Optimizer: Adam (adaptive learning optimizer)
- Loss Function: Binary Crossentropy
- Metrics: Accuracy

The model summary showed around ~100K trainable parameters (depending on vocabulary size and embedding dimension).

4. Model Training

- Trained for **10 epochs** with **batch size = 2**.
- Used **validation_split = 0.2** to monitor generalization performance.
- Observed steady improvement in both training and validation accuracy.

5. Model Evaluation

- The model achieved **high accuracy** on both training and validation sets.
- It effectively captured **semantic relationships** due to the bidirectional context awareness.

6. Visualization

- Plotted accuracy and loss curves across epochs.
- Observed convergence and stability, indicating good learning behavior.

Reason for Choosing Bidirectional LSTM

I selected **Bidirectional LSTM** because:

- Traditional LSTMs process data **only forward in time**, losing information about future words or context.
- **BiLSTMs read sequences in both directions**, providing a **richer understanding of context** — crucial for tasks like text classification, sentiment analysis, and sequence labeling.
- It's especially beneficial when the **meaning of a word depends on future words**, which happens often in natural language.

Conclusion

- The **Bidirectional LSTM** model demonstrated **better context understanding** compared to a simple LSTM.
- It achieved **high accuracy and stable convergence**, validating its effectiveness for sequence-based learning tasks.
- The model can be further enhanced by using **larger datasets, pre-trained embeddings (like GloVe or Word2Vec)**, and additional regularization techniques.

Final Outcome

The experiment successfully implemented a **Bidirectional LSTM neural network** capable of learning from both directions of a sequence, resulting in improved performance for natural language understanding.